

The Reconstructed Theory Has One Mass: a Machine-Checked Volume-Uniform Spectral Gap with Exact Identification Against the Gibbs Sums

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decision

Abstract

For the spatial $\mathbb{Z}_2$ (Ising-slice) system in the Dobrushin window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$, we machine-check in Lean 4 an end-to-end chain: (i) the Osterwalder–Schrader (site-form) reconstruction of the transfer operator is unitarily conjugate — by the explicit $\sqrt{w}$ boundary dressing — to the symmetrised Dobrushin kernel; (ii) the *unnormalised Gibbs sums themselves* are exact matrix elements of the reconstructed operator’s powers, $\text{gibbsPathSum } w \beta N A B = \lambda^N \langle \hat{T}^N Q A, Q B \rangle$ and the partition function is the same shape at the dressed constant — identities, not bounds; and (iii) there is *one* mass $m > 0$ such that for *every* spatial extent L the projected operator norm is at most e^{-m} and every *mixed* connected correlator of the reconstructed operator obeys $|\langle u, \hat{T}^n v \rangle - \langle \Omega, u \rangle \langle \Omega, v \rangle| \leq \|u\| \|v\| (e^{-m})^n$, the zero-time case included — constants per-observable, rate alone uniform in the volume. All quantifiers live in the Lean statements; the prose here copies the kernel’s cut. To situate the claim in the 2026 landscape of machine-checked constructive field theory: the abstract construction (reflection-positive data $\rightarrow$ physical Hilbert space $\rightarrow$ self-adjoint contractive transfer operator) has been machine-checked, sorry-free, in the Harvard formalization programme, *with the spectral gap carried as an undischarged hypothesis* (**GappedTransfer**); the Osterwalder–Schrader axioms have been machine-checked for the free field; and measure-side Dobrushin clustering has been machine-checked for lattice Yang–Mills. What we add, and what to our knowledge no machine-checked artefact yet contains, is the combination {concrete interacting model, gap *proved* rather than hypothesised, uniform in the spatial volume} on the reconstructed operator, *identified exactly against the raw Gibbs sums* — i.e. a concrete discharge of exactly the hypothesis shape the abstract frameworks carry. We do not claim Wightman reconstruction, a continuum limit, $SU(N)$, or any progress on the Clay problem.

1 Introduction and landscape

The Osterwalder–Schrader route builds a physical theory out of Euclidean data: a reflection-positive form gives a Hilbert space, and composition with the time shift gives a self-adjoint transfer operator. For a lattice slice system the construction is elementary — the content is

*With machine-checked artefacts produced on the programme’s Colab plane; toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned to `0764272048...`. Repository: THE-ERIKSSON-PROGRAMME, branch `d3-closure`, module anchor commit `3b0dfa9e81835106b6d43a1d582b8c12dcb8f36b`; verification anchor **[SLOT: fresh-clone SHA, pending pass 12]**.

never the construction but the *spectrum*: a spectral gap at every fixed spatial extent L is classical (Perron–Frobenius; in this programme it is the earlier `coupled_gap_all_eigenvalues`), and it degenerates as $L \rightarrow \infty$ unless something uniform survives. What a thermodynamic limit can actually consume is *one* $m > 0$ valid for every L at once.

That uniform input exists in this repository: the Dobrushin corollary `dobrushin_ising_uniform_gap` produces, inside the window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$, a single $m > 0$ bounding the projected transfer operator of the tilted kernel at every extent. This paper’s contribution is the *identification*, in two stages. Stage one (Section 3): the operator that bound governs *is* — unitarily, by the explicit $\sqrt{w}$ boundary dressing, with no spectral-transport theory — the transfer operator forced on the reconstructed (site-form) theory. Stage two (Section 5): the *unnormalised Gibbs sums themselves* are exact matrix elements of that operator’s powers — λ^N times an inner product, an identity valid at every real β and every positive weight, with the window entering only when the gap is invoked. The theorems are deliberately cheap: the same witnesses as D-6, no new estimate. Cheapness is the honesty of the claim — the uniformity is real because it is D-6’s, and the reconstruction adds commuting squares, machine-checked.

1.1 Machine-checked prior work, cited before a referee does

The following table is load-bearing and its rows were collected by a four-modality search on 2026-08-04 (evidence: `docs/OS-R-PRIORITY-COLLATION.md` in the repository; the four raw agent reports are appended there verbatim).

Artefact	What is machine-checked	What is not
mrdouglasny/ reflection-positivity (sorry-free ~2026-06)	RP data $\rightarrow$ Hilbert space (completion/quotient) $\rightarrow$ self-adjoint contraction T ; the dictionary $\langle [f], T^n[g] \rangle = \int f(g \circ \tau^n \circ \theta) d\mu$; trace formulas; susceptibility bounds <i>from</i> a gap	the gap itself: carried as the GappedTransfer <i>hypothesis</i> (GroundGap.lean); no spatial-volume-uniform statement; concrete instance is finite-lattice RP only
OSforGFF (arXiv:2603.15770)	all five GJ/OS axioms for the $d=4$ Gaussian free field, 0 sorries	reconstruction (listed as next milestone); any interacting model
xiyin137/OSreconstruction	Wightman-side GNS lane of the full OS $\leftrightarrow$ Wightman theorem	the theorem: 53 sorries + 12 axioms at last README snapshot
mrdouglasny/lgt (complete 2026-05-04)	Chatterjee-style Dobrushin uniqueness for lattice YM, $U(n)$, with exponential clustering whose constants depend only on (n, d, β)	an operator gap; any reconstruction; the statement is per-volume on the measure side
pphi2	transfer-matrix spectral machinery for φ_2^4	rests on 27–29 project axioms
This paper	the $\sqrt{w}$ -dressing unitary; the transported volume-uniform gap; the <i>exact</i> identities $\text{gibbsPathSum} = \lambda^N \langle \hat{T}^N QA, QB \rangle$ and $Z_N = \lambda^N \langle \hat{T}^N Q\mathbf{1}, Q\mathbf{1} \rangle$; volume-uniform <i>mixed</i> clustering with the zero-time case (os_reconstruction_measure_uniform ; core [SLOT: 8481, pass-11 measurement in flight] , oracle [SLOT: 3026, same] , 0 sorry)	Wightman axioms; continuum limit; $SU(N)$; Clay; the division-safe <i>normalised</i> six-term connected bound (named open step, Section 9)

There is a coordinated, active programme (Douglas et al., Harvard CMSA) whose declared live fronts are the OS reconstruction theorem, $P(\varphi)_2$, and YM mass gap at strong coupling, and whose framework carries the spectral gap as an undischarged hypothesis. The endpoint below is a concrete discharge of that hypothesis *shape* (different framework, same mathematical hole), which is why the comparison table above is printed in full before any claim wording.

2 The two geometries and the forced operator

Everything in this section is prior work of the repository, cited by module (the public trace of this lane is the ai.viXra 2607.009x cluster); no new mathematics appears here. The slice system carries two pairings on complex boundary vectors: the *site* form, which weighs by $1/w$ (**siteForm**, module **SpatialReconstruction**), and the *bond* form carrying the decoupled kernel

`spatialKernel` β (`bondForm`). The transfer operator is *forced*: $T = D_w K$ is the unique operator with $\text{site}(u, Tv) = \text{bond}(u, v)$ (`transferOp_unique`), self-adjoint for the site form at every β (`transferOp_selfAdjoint`) and positive for $\beta \geq 0$; reflection positivity of the underlying measure is the lane's earlier bond/site pair (modules `SpatialReflection`, `SpatialOS`). On the measure side, the same repository carries the Gibbs bridge (module `SpatialGibbs`): the unnormalised two-point path sum `gibbsPathSum` and partition function `gibbsPartition` of the coupled slice system, with the in-tree operator bridge `gibbsPathSum_eq_iterate` (the path sum as an N -fold kernel iterate against dressed observables, `dress` $w A = \sqrt{w} A$). On the Dobrushin side it carries `symWeighted` $= \sqrt{w} K \sqrt{w}$ (module `PerronGap`), its normalisation `tiltKernel` $= \text{symWeighted}/\lambda$ (module `DobrushinTilt`), the Euclidean packaging `opOf`, `vacOf`, `vacuumTransfer_opOf` (module `DobrushinTransport`), and the corollary `dobrushin_ising_uniform_gap` (module `DobrushinCorollary`). The seam fact this paper depends on — one `spatialKernel` declaration serving both geometries — is enforced by the elaborator in the module of Section 3.

3 The dressing unitary

All statements of this and the following sections live in one module (`YangMills/OS/OSReconstructionUniform.l` sha256 d9919e256ae59ac9...27bac149, 20,428 bytes, ten-pass ladder on the plane, final `ELAB_EXIT` 0). Write $Qu := \sqrt{w} u$ for the boundary dressing of a real Euclidean vector (`qEmbed`; it coincides with the measure side's `dress`, complexified — `qEmbed_eq_dress` is definitional); the site form weighs by $1/w$.

Theorem 3.1 (A1, `siteForm_qEmbed`). *For $w > 0$ and real u, v : $\text{siteForm } w (Qu) (Qv) = \sum_{\sigma} u_{\sigma} v_{\sigma}$. Per site the cancellation is elementary: $\sqrt{w} u \cdot \sqrt{w} v \cdot (1/w) = uv$.*

Theorem 3.2 (A2, `transferOp_qEmbed`). *$T(Qu) = Q(\text{act}(\text{symWeighted } w \beta) u)$ pointwise: the forced transfer operator on a dressed vector is the dressing of the symmetrised kernel's action.*

Theorem 3.3 (A3, `transferOp_iterate_qEmbed`). *$T^{[n]}(Qu) = Q((\text{act } S)^{[n]} u)$ — the shape the in-tree measure bridge `gibbsPathSum_eq_iterate` consumes.*

Together with A1, the dressing is a unitary between the Euclidean ℓ^2 pairing and the site form at every finite extent (the site form with $w > 0$ is definite, so no quotient intervenes — this is the elementary, finite-volume case of the OS construction, and we say so plainly).

4 The transported volume-uniform gap

Theorem 4.1 (`os_reconstruction_uniform_gap`). *Let $\beta, \gamma \in \mathbb{R}$ and $0 < \alpha < 1$ with $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha$. There is $m > 0$ such that for every L there are $\lambda > 0$ and $\Omega > 0$ with the fixed-point equation $\sum_{\tau} \text{tiltKernel}(\text{sliceW } \gamma L) \beta \lambda \sigma \tau \Omega_{\tau} = \Omega_{\sigma}$, the projected bound $\|\text{projectedTransfer}(\text{opOf}(\text{tiltKernel}) e^{-m})\|$, and the A2 identity at $w := \text{sliceW } \gamma L$.*

The proof is an obtain on `dobrushin_ising_uniform_gap` followed by A2: *the same witnesses*, no new analytic content — which is precisely the point. The uniformity is real because it is D-6's; the reconstruction contributes the identification, not a new estimate.

5 Exact identification against the Gibbs sums

The bridge from the raw measure to the operator is a chain of *identities* — no bound appears until the gap is invoked, and none of the statements below needs the window: they hold at every real β , every $\lambda \neq 0$, every positive weight.

Lemma 5.1 (B0b, `act_symWeighted_eq_smul_act_tilt`). *act* $Su = \lambda \cdot \text{act}(\text{tilt})u$ for $\lambda \neq 0$, and iterating (`act_symWeighted_iterate_eq_smul_tilt`, via the scalar transport lemmas `act_smul_fun`, `act_iterate_smul_fun`): $(\text{act } S)^{[N]}u = \lambda^N \cdot (\text{act tilt})^{[N]}u$.

Lemma 5.2 (B0a, `act_iterate_eq_opOf_pow`). $(\text{act } M)^{[n]}u = (\text{opOf } M)^n(\text{toLp } u)$ pointwise, for any kernel matrix M .

Theorem 5.3 (B1, `gibbsPathSum_eq_inner_pow`). For $w > 0$, $\beta \in \mathbb{R}$, $\lambda \neq 0$, $N \in \mathbb{N}$ and real observables A, B :

$$\text{gibbsPathSum } w \beta N A B = \lambda^N \langle (\text{opOf } (\text{tiltKernel } w \beta \lambda))^N (QA), QB \rangle,$$

an exact identity: the unnormalised Gibbs two-point sum is λ^N times a matrix element of the reconstructed operator's powers, taken between dressed observables.

Theorem 5.4 (B2, `gibbsPartition_eq_inner_pow`). Under the same hypotheses, the partition function is the same shape at the dressed constant: $\text{gibbsPartition } w \beta N = \lambda^N \langle \hat{T}^N(Q\mathbf{1}), Q\mathbf{1} \rangle$.

Remark 5.5. Dividing B1 by B2 cancels λ^N , so the normalised two-point function equals $\langle \hat{T}^N QA, QB \rangle / \langle \hat{T}^N Q\mathbf{1}, Q\mathbf{1} \rangle$ — plain arithmetic from the two identities, recorded here as prose. The quantitative six-term connected bound for this quotient (division-safe, with an explicit floor for the denominator) is the one named open step of the campaign; see Section 9. We print the boundary between identity and estimate exactly where the kernel puts it.

6 Mixed clustering on a generic Hilbert space

The clustering machinery is stated for an arbitrary real Hilbert space H (a `section` with `[NormedAddCommGroup H]` `[InnerProductSpace ℝ H]` `[CompleteSpace H]`), so the three lemmas below are reusable against any transfer operator packaged as a `VacuumTransfer` — in particular they are exactly the consumer shape of the `GappedTransfer` hypothesis of the abstract frameworks in Section 1.1.

Lemma 6.1 (`norm_sub_inner_smul_le`). For $\|\Omega\| = 1$: $\|v - \langle \Omega, v \rangle \Omega\| \leq \|v\|$ (Pythagoras; removing the vacuum component does not grow the norm).

Lemma 6.2 (`pow_apply_norm_le`). $\|S^n v\| \leq r^n \|v\|$ whenever $\|S\| \leq r$, $r \geq 0$.

Theorem 6.3 (`mixed_connCorr_bound`). Let T be a `VacuumTransfer` with vacuum Ω , and suppose $\|\text{projectedTransfer } T \Omega\| \leq r$ with $r \geq 0$. Then for all $u, v \in H$ and all $n \in \mathbb{N}$ including $n = 0$:

$$|\langle u, T^n v \rangle - \langle \Omega, u \rangle \langle \Omega, v \rangle| \leq \|u\| \|v\| r^n.$$

The $n = 0$ case is the Pythagoras lemma; $n \geq 1$ goes through the in-tree `projected_pow_succ`.

This strictly extends the lane's earlier diagonal clustering (`os_reconstruction_uniform_clustering`, $u = v$, constant $\|v\|^2$), which remains in the module: two *different* observables, and the time-zero correlator is bounded too — the zero-time case is where a naive geometric-series argument is silent, and it is carried by the projection geometry instead.

7 The endpoint

Theorem 7.1 (`os_reconstruction_measure_uniform`). *Let $\beta, \gamma \in \mathbb{R}$ and $0 < \alpha < 1$ with $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha$. There is one $m > 0$ such that for every spatial extent L there are $\lambda > 0$ and $\Omega > 0$ with:*

- (i) *the fixed-point equation for `tiltKernel (sliceW  $\gamma$  L)  $\beta$   $\lambda$` at Ω ;*
- (ii) *exact identification: for every N and all real observables A, B , `gibbsPathSum (sliceW  $\gamma$  L)  $\beta$   $N$  A B =  $\lambda^N \langle \hat{T}^N(QA), QB \rangle$` with $\hat{T} = \text{opOf}(\text{tiltKernel} \dots)$;*
- (iii) *volume-uniform mixed decay: for all u, v in the slice Hilbert space and all $n \geq 0$, $|\langle u, \hat{T}^n v \rangle - \langle \text{vacOf } \Omega, u \rangle \langle \text{vacOf } \Omega, v \rangle| \leq \|u\| \|v\| (e^{-m})^n$.*

The witnesses of (i)–(iii) are `dobrushin_ising_uniform_gap`'s, consumed verbatim; the packaging is `vacuumTransfer_opOf + tiltKernel_symm`; (ii) is B1 at $w = \text{sliceW } \gamma L$ (whose positivity is the in-tree `sliceW_pos`); (iii) is Theorem `mixed_connCorr_bound` at $r = e^{-m}$. One theorem, three clauses, no new constant: the mass that clause (iii) exhibits is the mass D-6 proved, now measured against the raw Gibbs sums through the exact identity of clause (ii).

8 Reproduction

All numbers below are read from committed run artefacts; predictions were committed *before* their measurements (the programme's standing discipline — the ledger records nine consecutive exact count predictions for this lane before this paper's two).

Stage OS-R-1 (measured, ledger Addenda 616–619)

Quantity	Predicted	Measured
core jobs, unwired	8480 (ledger baseline)	8480
core jobs, wired	$8480 + 1$	8481
oracle reports, wired	$3010 + 7$	3017
<code>sorryAx</code> occurrences	0	0
endpoint axiom sets	standard triple	exactly [<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>]

Fresh-clone witness at anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`: judges 110/110 in normal and -0; core 8481; oracle 3017; module sha256 bit-identical (Addendum 619; run `cell scripts/osr1_pass7_freshclone.py`).

Stage OS-R-2 (predictions registered, Addendum 621; measurements [SLOT: pass-11/12 in flight])

Quantity	Predicted (committed first)	Measured
core jobs	8481 exact (file grew, no module added)	[SLOT: pass 11]
oracle reports	$3017 + 9 = 3026$ exact	[SLOT: pass 11]
<code>sorryAx</code> occurrences	0	[SLOT: pass 11]
fresh-clone reproduction	bit-identical module, same counts	[SLOT: pass 12]

Module: `YangMills/OS/OSReconstructionUniform.lean`, sha256 `d9919e256ae59ac9189e397e5abf1249cfaca32` (20,428 bytes), banked at commit `3b0dfa9e8` after the plane printed `ELAB_EXIT 0` on byte-identical source (the cell prints size and sha256 before elaborating; both matched the local file before banking). Pass history: passes 1–5 (OS-R-1, Addenda 614–616), passes 8–10 (OS-R-2, Addendum 621: the inner-product notation trap at this Mathlib pin, then a single `nlinarith` orientation failure in the Pythagoras lemma, then green). The wiring unit with the registered predictions is committed as `scripts/osr2_pass11_wiring.py` *before* its run, at commit `520e177a7`.

9 What is not claimed

No Wightman axioms, no $OS \rightarrow$ Wightman analytic continuation, no continuum limit, no $SU(N)$, no statement about the Clay problem. The window is the Dobrushin window; outside it the programme’s own kill-test (repository, Addendum 610) shows the standard-cone Birkhoff route is projectively blind to the site weight, which is why no claim beyond the window appears here. The model is the $\mathbb{Z}_2$ Ising slice; “reconstruction” means the elementary finite-volume site-form construction (definite form, no quotient), and the theorems are about its *spectrum and its exact relation to the Gibbs sums*, not about the construction being difficult. One step of the campaign charter remains open and is named rather than blurred: the division-safe *normalised* six-term connected bound — the quantitative decay of `gibbsPathSum/Z`— (product of one-point quotients), with an explicit denominator floor. Its ingredients are exactly B1, B2 and Theorem `mixed_connCorr_bound`; nothing in Sections 3–7 depends on it, and the endpoint theorem stands on its own.

References

- [1] M. R. Douglas, S. Hoback, C. Mei, Y. Nissim, *Formalization of QFT*, arXiv:2603.15770 (2026). Repositories: `mrddouglasny/OSforGFF`, `mrddouglasny/reflection-positivity`, `mrddouglasny/lgt`, `mrddouglasny/pphi2`; `xiyin137/OSreconstruction`.
- [2] K. Osterwalder, R. Schrader, *Axioms for Euclidean Green’s functions* I, II, Comm. Math. Phys. **31** (1973), **42** (1975).
- [3] J. Glimm, A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer (1987).
- [4] R. L. Dobrushin, *The description of a random field by means of conditional probabilities and conditions of its regularity*, Theory Probab. Appl. **13** (1968).
- [5] L. Eriksson, *From the Gibbs Weight to the Spectral Gap: A Complete Machine-Checked Osterwalder–Seiler Chain for the $\mathbb{Z}_2$ Lattice Gauge Chain*, ai.viXra:2607.0078 (2026) — the zero-extension, explicitly non-volume-uniform predecessor of this paper.
- [6] L. Eriksson, *the dobrushin-matrix paper of the same repository* (`papers/dobrushin-matrix/`), source of `dobrushin_ising_uniform_gap` (D-6, ledger Addendum 603).